■ Remontada Ghost Ball EV

Spatial Physics & Pitch Control Engine

Track. Project. Exploit. — A 60+ FPS vectorized off-ball Expected Value (EV) and ray-casting tactical engine.

Phase 2 of the Remontada Ecosystem. This repository contains the source code for the spatial-physics and pitch-control layer. For complete mathematical derivations—including 2D homography matrices, non-linear xT topology, biomechanical turnaround penalties, vectorized ray-cast interference, and EMA smoothing—refer to the Master Repository.

■ Execution & Rendering Flow

1	■ Input Broadcast Video	test_clip_1.mp4 or Synthetic
2	■ YOLOv8 + ByteTrack	Detection & tracking via main.py
3	■ Pitch Mapping	HSV K-Means team clustering + 2D homography
4	■ Pipeline Adapter	RemontadaPipelineAdapter
5	■ EV Physics Engine	7,072-cell SIMD pitch-control / xT / ray-cast
6	■■ Broadcast Overlay	Ghost Ball + Laser Tripwire
7	■ H.264 Export	remontada_annotated_output.mp4

■ File Structure

File	Purpose
main.py	Entry point: YOLOv8m + ByteTrack, homography, K-Means, H.264 exporter
pipeline_integration.py	Adapter bridging raw tracker dictionaries into contiguous NumPy arrays
remontada_engine.py	Pure NumPy vectorized physics engine: pitch control, xT, ray-casting
remontada_overlay.py	Broadcast AR visualizer: sonar ripples, Ghost Ball, shatter tripwire
README.md	Quick-start documentation

■ Quick Start & CLI Usage

1. Install Dependencies

The pipeline includes a pre-flight dependency checker and uses **imageio-ffmpeg** to provide cross-platform H.264 MP4 encoding without relying on potentially broken OpenCV Windows codecs.

```
pip install numpy opencv-python torch ultralytics scipy imageio imageio-ffmpeg
```

2. Run the Engine — Zero-Config Synthetic Fallback

If no match clip is present in the directory, running **main.py** automatically synthesizes a 350-frame, 1280 × 720, 25 FPS broadcast-style match video and runs the full YOLOv8 + EV pipeline.

```
python main.py
```

3. Run on Custom Match Footage

Pass a broadcast **.mp4** clip and customize the attacking direction or frame limit using CLI flags.

```
python main.py --video match_clip.mp4 --output tactical_audit.mp4 --model yolov8m.pt
--attacking-team 1 --max-frames 500
```

CLI Arguments

Argument	Default	Description
--video	test_clip_1.mp4	Input match video. Automatically generates a synthetic clip if missing.
--output	remontada_annotated_output.mp4	Output H.264 (yuv420p) annotated video.
--model	yolov8m.pt	Ultralytics YOLO weight file used for player/ball detection.
--attacking-team	1	Team ID (0 or 1) attacking toward X = 104 m.
--max-frames	None	Optional frame-processing cap for rapid previews.
--no-display	False	Disables the live cv2.imshow('Remontada Live') window for headless runs.

■ Live Broadcast HUD Legend

Cyan / Crimson Corner Brackets & Grass Ripples

Active player bounding boxes and foot-anchored sonar ellipses scaled dynamically by local Pitch Control dominance.

Pulsing White / Emerald "Ghost Ball" Orb

The exact (X, Y) pitch coordinate with the highest smoothed Expected Value (EV) for an immediate pass.

Crimson Laser + Shatter Effect — BLOCKED

Triggers when a defender's interception radius (1.15 m) physically blocks the linear ray-cast lane between the ball carrier and the Ghost Ball pocket.

■ Engine Overview

The engine combines broadcast-video perception with a spatial tactical model. Player and ball detections are transformed from image coordinates into pitch coordinates using homography, then processed as contiguous NumPy arrays for vectorized tactical calculations. The EV layer evaluates candidate pitch locations, while ray-casting estimates whether defensive presence can physically interfere with the proposed passing lane.

Layer	Role
Perception	YOLOv8 + ByteTrack for player and ball detection/tracking.
Team Identification	HSV feature extraction + K-Means clustering.
Spatial Mapping	2D pitch homography converts image coordinates to metric pitch coordinates.
Physics / EV	Vectorized pitch control, expected-value topology, and ray-cast interception logic.
Temporal Stability	EMA smoothing reduces frame-to-frame noise in tactical outputs.
Visualization	Ghost Ball, sonar-style control ripples, and blocked-lane tripwire effects.

Master Repository: Replace LINK_TO_YOUR_MASTER_REPO_HERE in the source README with the URL of the main Remontada repository before publishing.

Project Tagline

Track. Project. Exploit.
A spatial-physics engine for turning broadcast football footage into continuous tactical EV signals.